# Dazaet Galicia-Badillo

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10 October 2026

Dr Daniel Lee
Department of Chemical Engineering, School of Engineering
The University of Manchester
Manchester M13 9PL, United Kingdom

**Re: Research Associate in Solid-State NMR Spectroscopy, ref. SAE-032399**

Dear Dr Lee,

I am writing to apply for the position of Research Associate in Solid-State NMR Spectroscopy (ref. SAE-032399) within the EPSRC Programme Grant on Molecular Solid Solutions. I completed my PhD in Chemical Sciences at the Institute of Chemistry, UNAM (Mexico), in 2026 under the supervision of Prof. Braulio Rodríguez-Molina, and the degree was awarded with honours.

Solid-state NMR was the central technique of my PhD. On a Bruker 500 MHz spectrometer with a double-resonance ($^{1}$H/X) MAS probe, I set up my own experiments, including tuning and matching both channels and retuning the $^{1}$H channel for $^{19}$F. I acquired $^{13}$C CP-MAS spectra with variable contact times, $^{19}$F MAS spectra, variable-temperature $^{2}$H spectra and some $^{15}$N CP-MAS spectra. I simulated $^{2}$H lineshapes with NMR WebLab to obtain rotational rates and activation energies, and analysed $^{13}$C $T_1$ and $^{1}$H and $^{19}$F relaxometry data to follow faster motions.

I use NMR together with diffraction. In a pyrrolo[3,2-b]pyrrole solvate that jumps on heating (Mater. Chem. Front., 2026), $^{2}$H NMR and variable-temperature single-crystal diffraction allowed me to separate the molecular dynamics from the anisotropic thermal expansion that drives the jump. In a zirconium MOF (ACS Mater. Lett., 2026), the same combination showed how a photoswitch and guest uptake change the motion of a molecular rotor. Both projects were done with Dr Martin Dračínský and Dr Jan Blahut at IOCB Prague.

The role also involves preparing the solid solutions, and synthesis and crystallisation are a large part of my training. I grow single crystals of molecular crystals, cocrystals and solvates and characterise them by SCXRD, PXRD, DSC, TGA and hot-stage microscopy. I contributed to a study in which the salient temperature of cocrystals was tuned through the composition of the crystallisation solvent mixture (Angew. Chem. Int. Ed., 2026). I also run DFT (Gaussian16, periodic CP2K) and molecular dynamics (LAMMPS) on HPC clusters. Diffraction gives the average structure of a molecular solid solution, while solid-state NMR can show how a dopant is distributed and how it moves locally, and this is the question I would like to work on within MoSS.

I have not yet used DNP. The UK DNP MAS NMR Facility at Manchester, and the chance to extend my work to $^{17}$O and natural-abundance $^{15}$N, are among my main reasons for applying. During my PhD I learned periodic DFT and molecular dynamics from scratch and now use both routinely, and I would approach DNP in the same way.

My 14 publications include collaborations with groups in the Czech Republic, the United States, China and the United Arab Emirates. I co-supervised four bachelor's theses and nine undergraduate research placements, trained students in laboratory procedures and safety, and handled purchasing and inventory for our laboratory. I have presented 15 posters and 9 talks, and received the Best Poster Award at DynaMIC23 in Fribourg. Thank you for considering my application; I would be glad to discuss it at interview.

Yours sincerely,

**Dazaet Galicia-Badillo, PhD**



# Dazaet Galicia-Badillo

PhD Chemical Sciences
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## About me

My research has been focused on the design and synthesis of organic crystals and metal-organic frameworks exhibiting dynamics from the molecular level to the macroscopic level, including rotational motion, photoisomerization, crystal flexibility, salient mechanical responses, and single-crystal-to-single-crystal phase transitions. I bring a strong experimental toolkit: organic synthesis, multinuclear solid-state NMR ($^{13}$C, $^{19}$F, $^{31}$P, $^{2}$H, $^{15}$N), single-crystal and powder X-ray diffraction (SCXRD/PXRD), fluorimetry, and DFT calculations. My long-term goal is to build an independent research program that bridges materials chemistry with applied challenges in energy storage, environmental remediation, or biomedical applications in academia or the private sector.

## Honors, Awards & Funding

Nominated* Alfonso Caso Medal nomination - UNAM. 2026
Nominated for recognition of outstanding postgraduate achievement (*outcome pending).

Ph.D. awarded with honors Chemical Sciences - UNAM.
2026

Best Presentation Award - IX Seminario Regional de Materiales Avanzados, UAEH, Mexico, October
2025

Supercomputing resource allocations – National High-Performance Computing Laboratory Search (LANCAD)
2024
Prepared successful proposals securing two allocations of computational resources for research.

Best Poster Award - DynaMIC23, Fribourg, Switzerland
2023
Dynamic Materials, Crystals and Phenomena Conference hosted by Prof. Simon Krause

Doctoral scholarship - Secretariat of Science, Humanities, Technology, and Innovation (SECIHTI) 2022
Support for Ph.D. studies

Master's scholarship - Secretariat of Science, Humanities, Technology, and Innovation (SECIHTI) 2020
Support for Master's studies

Institute of Chemistry Undergraduate Scholarship -UNAM
2019
Awarded for the best undergraduate research project at the Institute of Chemistry

## Education

**PhD Chemical Sciences - UNAM, Institute of Chemistry** 2022 - 2026
Thesis: Molecular and Macroscopic Motion in Organic Crystals and Metal-Organic Frameworks
Advisor: Prof. Dr. Braulio Rodríguez-Molina

**M.Sc. in Chemical Sciences - UNAM, Institute of Chemistry** 2020 - 2022

**B.S. in Chemistry - UNAM, Faculty of Chemistry** 2015 - 2020

## Technical skills

**Experimental**: Organic synthesis, including air- and moisture-sensitive reactions (Schlenk techniques); MOF solvothermal synthesis and post-synthetic modification (PSM). Solid-state NMR (Bruker 500 MHz, double-resonance $^{1}$H/X MAS probe): $^{13}$C CP-MAS, $^{19}$F MAS, $^{31}$P, variable-temperature $^{2}$H and $^{15}$N CP-MAS; $^{2}$H lineshape simulation; DFT calculation of NMR parameters (CASTEP); analysis of $^{13}$C $T_1$ and solid-state $^{1}$H and $^{19}$F relaxometry data. X-ray diffraction: SCXRD and PXRD, including variable-temperature studies and structure refinement (ShelX). Photophysics: steady-state and time-resolved fluorescence, UV–Vis absorption. Thermal analysis: hot-stage microscopy, DSC, TGA. **Computational & Software:** DFT calculations (geometry optimization, electronic structure, energy profiles, PES) in Gaussian16; periodic DFT in CP2K and molecular dynamics simulations (LAMMPS) on HPC clusters. Crystallography tools: ShelX, CrystalDiffract, CCDC suite. Visualization: CrystalMaker, ChemDraw. NMR processing and simulation: TopSpin, MestreNova, NMR WebLab, CASTEP. AI-assisted workflows for research automation and data

analysis. **Lab management:** Purchasing and vendor management; inventory control and lab organization; team leadership and training; funding acquisition (grants/proposals).

## PUBLICATIONS (RESEARCH, REVIEW AND PREVIEW)

14. Ernesto Enriquez-Palacios, **Dazaet Galicia-Badillo**, Marcos Flores-Alamo, Jose Luis Belmonte-Vazquez. Imidazole-based Photoswitches by Design and their use for Data Encryption. *ACS Applied Materials & Interfaces*. **2026.** 18, 35, 47895–47906. *[IF: 8.5]*

**13. Dazaet Galicia-Badillo**, Ernesto A. Hernández-Morales, Yesterday López-Herrera, Erick Hernández-Santiago, Jan Blahut, Rubén A. Toscano, Martin Dracínský, Braulio Rodríguez-Molina. Dissecting Molecular and Macroscopic Dynamics in a Jumping Solvate. *Materials Chemistry Frontiers,* **2026**. 10, 18, 3305-3313. *[IF: 8.1]*

**12. Dazaet Galicia-Badillo**, Jade Cisneros, Jan Blahut, Alonso Acosta-Vera, Alfonso García-Márquez, Haggeo Desirena, Erika Bartůnková, Ma. Carmen García-González, Ingrid Chávez-Serrano, Joelis Rodríguez-Hernández, Martin Dracínský, Braulio Rodríguez-Molina. Integration of a Molecular Rotor and a Photoswitch in a Metal-Organic Framework. *ACS Materials Letters*, **2026**. 8, 7, 1995-2001. *[IF: 8.7]*

**11.** Ernesto A. Hernández-Morales, Dayra Barreto-Hernández, **Dazaet Galicia-Badillo**, Rubén A. Toscano, M. Elena García-Aguilera, Braulio Rodríguez-Molina. Crystals as Rockets: Modulation of the Salient Temperature in Cocrystals by Solvent Mixture Composition. *Angewandte Chemie Int. Ed.*, **2026**, e26010. *[IF: 16.9]*

**10.** Ángel D. Álvarez-Castillo, Alexis Juárez-Morales, **Dazaet Galicia-Badillo**, Mauricio Maldonado-Domínguez, José M. Heredia-Peñaloza, Marcos Flores-Álamo, Jose Luis Belmonte-Vázquez. Rational Modulation of Excited-State Intramolecular Proton Transfer, Aggregation-Induced Emission, and Intramolecular Motion on Polysubstituted Imidazoles (Front Cover). *ChemPhotoChem*, **2026**, 10, e70099. *[IF: 3.0]*

**9.** Mario Castañón-García, Pedro López-Mendoza, **Dazaet Galicia-Badillo**, Luis D. Miranda. Synthesis of Non-Symmetrical 3,3'-Bicoumarins: Total Synthesis of Arteminorin C, 3,3'-Biisofraxidin and Biscopoletin. *Organic Letters*, **2025**, 27 (21), 5349-5354. *[IF: 5.0]*

**8.** José L. Belmonte-Vázquez, Durga Prasad Karothu, Carl H. Fleischer, **Dazaet Galicia-Badillo**, Mauricio Maldonado-Domínguez, Robert W. Schurko, Liang Li, Pance Naumov, Braulio Rodríguez-Molina. Low-Temperature Dynamics at Nano- and Macroscales: Organic Crystal That Exhibits Low-Temperature Molecular Motion and the Thermosalient Effect. *Chemistry of Materials*, **2025**, 37, 3373-3383. *[IF: 7.2]*

**7. Dazaet Galicia-Badillo**, Braulio Rodríguez-Molina. On the Flexibility of 1,4-Bis(phenylethynyl)benzene Butyl Ester Organic Crystal Featuring Mobile Components. *Tetrahedron*, **2024**, 167, 134245. *[IF: 2.2]*

**6. Dazaet Galicia-Badillo**, José L. Belmonte-Vázquez, Mario Rodríguez, Braulio Rodríguez-Molina, Ma. Carmen García-González. Aggregation-Induced Emission Enhancement and Solid-State Photoswitching of Crystalline Carbazole N-Salicylidene Anilines. *ACS Omega*, **2024**, 9 (36), 38015-38022. *[IF: 4.3]*

**5.** Alonso Acosta-Vera, **Dazaet Galicia-Badillo**, Armando Navarro-Huerta, Konstantin Stracke, Virginia Gómez-Vidales, Joelis Rodríguez-Hernández, Jack D. Evans, Braulio Rodríguez-Molina. Modulation of the Photophysics and Internal Dynamics in a Zr Metal-Organic Framework by the Inclusion of Fluorinated Guests. *ACS Materials Letters*, **2024**, 6 (9), 4395-4401. *[IF: 8.7]*

**4. Dazaet Galicia-Badillo**, Paulina Noemí García-Vargas, Braulio Rodríguez-Molina. Preview: The Integration of Multiple Components for Multi-Responsive Dynamic Framework Materials. *Matter*, **2024**, 7 (8), 2766-2768. *[IF: 17.5]*

**3.** Erick Hernández-Santiago, Jorge Espinosa-Rocha, **Dazaet Galicia-Badillo**, Braulio Rodríguez-Molina. Review: Local and Global Dynamics in Organic Soft Porous Crystals. *Matter*, **2024**, 7 (2), 408-429. *[IF: 17.5]*

**2.** Manuel Rodríguez-Molina, **Dazaet Galicia-Badillo**, Enoc Cetina-Mancilla, Jorge Cárdenas, Lilian I. Olvera, Rubén A. Toscano, Braulio Rodríguez-Molina, Mikhail G. Zolotukhin. 9-Trifluoromethylxanthenediols: Synthesis and Supramolecular Motifs. *ACS Omega*, **2022**, 7 (16), 13520-13528. *[IF: 4.1]*

**1.** Aaron Torres-Huerta, **Dazaet Galicia-Badillo**, Andrés Aguilar-Granda, Jacob T. Bryant, Fernando J. Uribe-Romo, Braulio Rodríguez-Molina. Multiple Rotational Rates in a Guest-Loaded, Amphidynamic Zirconia Metal-Organic Framework. *Chemical Science*, **2020**, 11 (42), 11579-11583. *[IF: 9.8]*

## TEACHING & MENTORSHIP

Co-director of 4 bachelor's theses (2023-present), covering MOF synthesis, organic crystal dynamics, and luminescent materials. Supervisor of 9 undergraduate short research stays (2022-2025) and 1 high school student.

## Scientific conferences & presentations

15 posters (6 international, 9 national) · 9 oral presentations (4 invited lectures, 5 congress talks)

## Scientific divulgation

1 article (Boletín IQ-UNAM, 2022) · 2 public lectures (High Schools, 2024–2025) · 1 media consultation (UNAM Global, 2022)

## References

**Prof. Dr. Braulio Rodríguez Molina** - PhD Supervisor**.** Instituto de Química, Universidad Nacional Autónoma de México
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